

Sophia Sharma

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 84% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: I2C/SPI/UART, PCB Debugging, C/C++, Embedded Linux, MQTT, FreeRTOS, ARM Cortex, Python

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Bosch* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core I2C/SPI/UART service, reducing latency by 34%.
- Led a squad of 7 engineers delivering scalable microservices processing 28M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 66%.

Associate Embedded Systems & IoT Firmware Engineer — *Qualcomm* (2019 - 2023)

- Designed and maintained PCB Debugging data endpoints with 96% SLA uptime.
- Optimized database indexing and caching layers using C/C++ saving \$38k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, PCB Debugging, C/C++, Embedded Linux]

Engineered end-to-end production solution leveraging I2C/SPI/UART and PCB Debugging with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [C/C++, Embedded Linux, MQTT, FreeRTOS]

Architected high-resilience system integrating C/C++ with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

Stanford University — 2015 - 2019 (Score: 9.4 / 10.0)